



Volumes

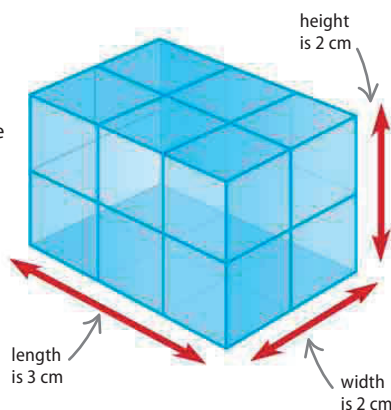
THE AMOUNT OF SPACE WITHIN A THREE-DIMENSIONAL SHAPE.

Solid space

When measuring volume, unit cubes, also called cubic units, are used, for example, cm^3 and m^3 . An exact number of unit cubes fits neatly into some types of three-dimensional shapes, also known as solids, such as a cube, but for most solids, for example, a cylinder, this is not the case. Formulas are used to find the volumes of solids. Finding the area of the base, or the cross section, of a solid is the key to finding its volume. Each solid has a different cross-section.

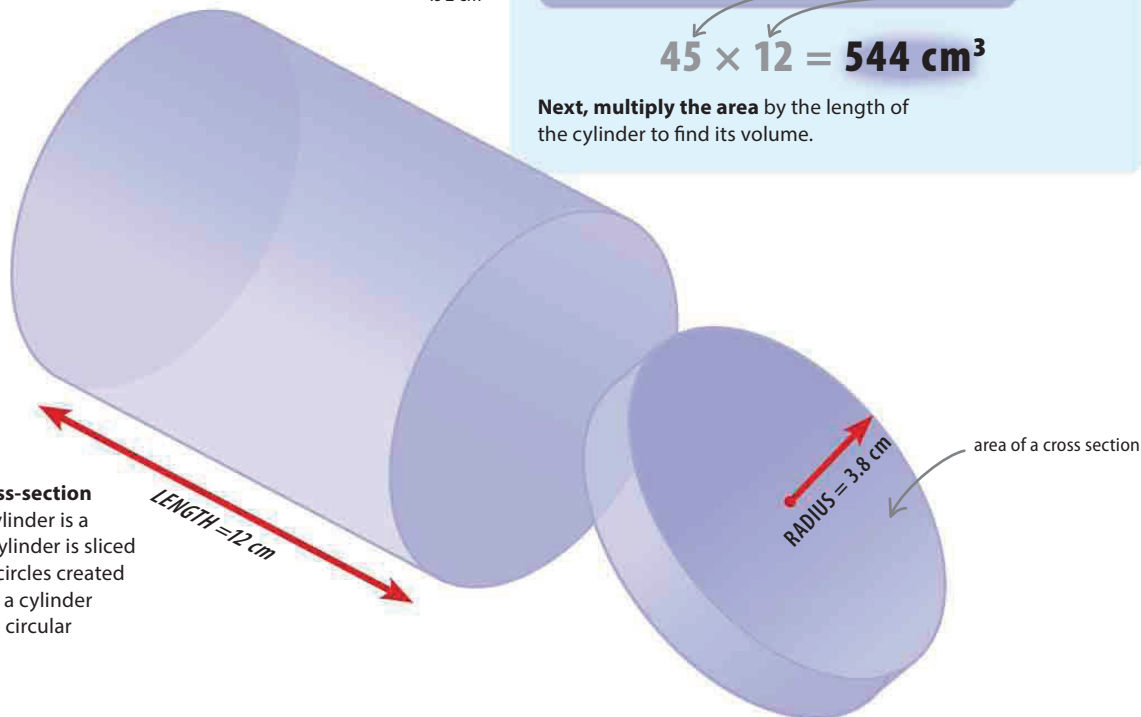
Unit cubes

A unit cube has sides that are of equal size. A 1 cm cube has a volume of $1 \times 1 \times 1$ cm, or 1 cm^3 . The space within a solid can be measured by the number of unit cubes that can fit inside. This cuboid has a volume of $3 \times 2 \times 2$ cm, or 12 cm^3 .



Circular cross-section

The base of a cylinder is a circle. When a cylinder is sliced widthwise, the circles created are identical, so a cylinder is said to have a circular cross-section.



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◀ 152–153 Solids

Surface area of solids 156–157 ▶

Finding the volume of a cylinder

A cylinder is made up from a rectangle and two circles. Its volume is found by multiplying the area of a circle with the length, or height, of the cylinder.

$$\text{volume} = \pi \times r^2 \times l$$

formula for finding volume of a cylinder

The formula for the volume of a cylinder uses the formula for the area of a circle multiplied by the length of the cylinder.

equals 3.14

$$\text{area} = \pi \times r^2$$

formula for finding area of a circle

$$3.14 \times 3.8 \times 3.8 = 45 \text{ cm}^2$$

area of cross section, given to 2 significant figures

First, find the area of the cylinder's cross-section using the formula for finding the area of a circle. Insert the values given on the illustration of the cylinder below.

$$\text{volume} = \text{area} \times \text{length}$$

$$45 \times 12 = 544 \text{ cm}^3$$

Next, multiply the area by the length of the cylinder to find its volume.